Experiment - 1

Aim :- Design an ER diagram for an indian e-Commerce platform with entities customer, product, order, orderitem, seller, category, payment, Delivery, Address. Specify primary key, Composite and multi-valued attributes, weak entities, Specialization of Product, and participation Constraints.

Theory :-

The ER (Entity - Relationship) model, given by Peter chen, is a Conceptual model used to design a database by representing real world real world data as entities, attributes, and relationships, independent of any DBMS.

Key terms :-

Entity - A real world object with independent existence (e.g. customer, product).

Attribute :- A property of an entity. Types Simple (atomic), Composite (divisible ; e.g., abc Name → first + last), Single valued ; multi-valued (Can take many values, Shown as double oval :), and key (uniquely identifies the entity ; underlined).

Relationship — An association between two or more instances of one entity relate to another.

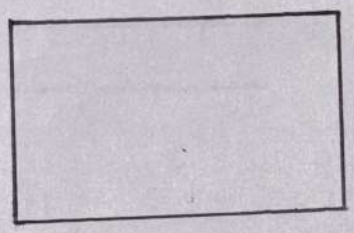
participation Constraint — Total (every instance must participate, Shown with a bold/ double line) or partial (optional, Single line).

Weak entity — An entity with no key of its own; identified only through its partial key Combined with

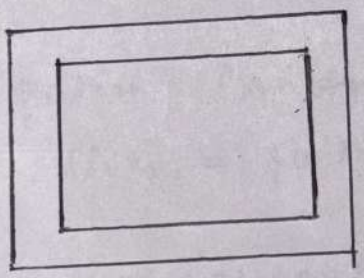
a related Strong (owner) entity, connected via double diamond identifying relationship (e.g. Order item depends on order).

Specialization / Generalization :- dividing an entity into subclasses with distinct attributes (Specialization) or combining entities into a superclass (Generalization), shown using an ISA triangle (e.g., product → electronics, fashions, Grocery ; Books).

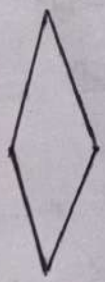
* Notation used :



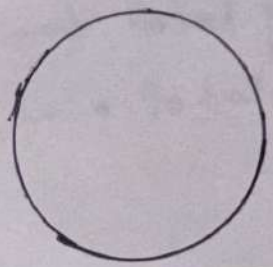
→ Strong entity



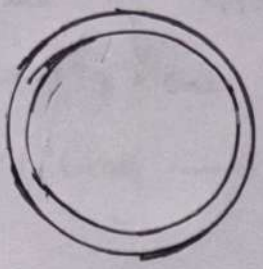
→ Weak entity



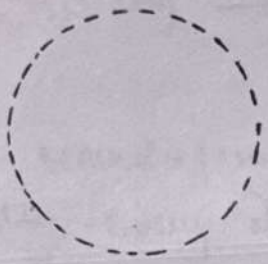
→ Relationship



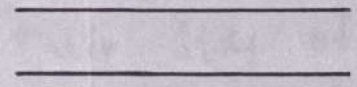
→ Attribute



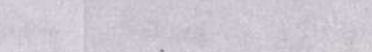
→ Multivalued attributes



→ Composite attribute



→ key attribute



→ partial key

- participation Constraints (bold/double lines = total, thin lines = partial, visible in the diagram):

Order → Total participation in places (an order must have a customer.

Address → Total participation in has an address. Can't exist with a customer).

order item → total participation in Constraints. and it refers to (an order item can't exist without order).

product → Total participation in sells and classifies.

payment → Total participation in Generates, and order

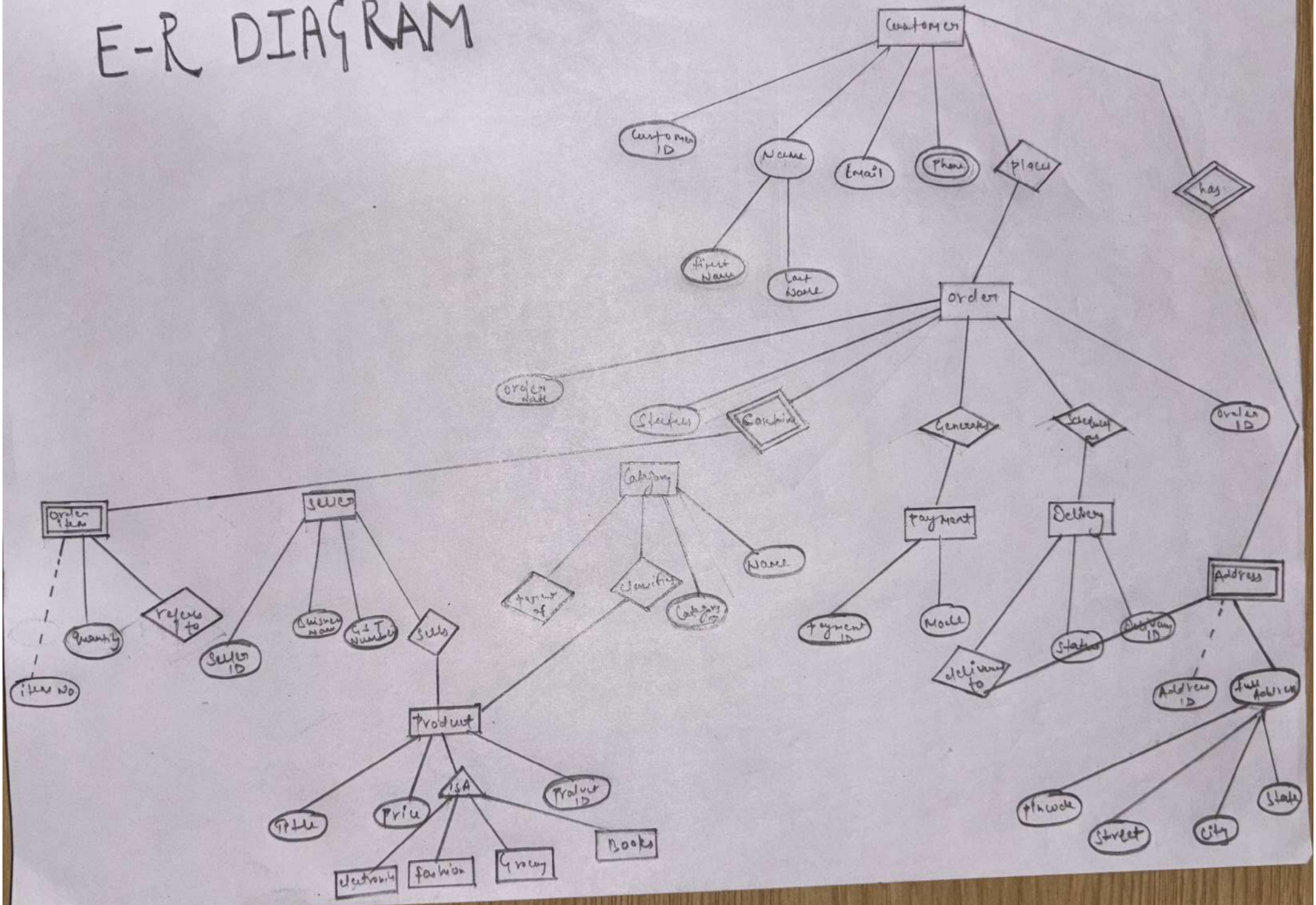
Side of Generates is also total (every order must have exactly one payment, and vice versa).

Delivery ⟶ total participation in Scheduled as, same logic – 1:1 total on both sides.

Delivery ⟶ partial participation in delivered to (this on the delivery side, since not shown as mandatory in the same strict sense, though logically every delivery needs an address).

Customer, Seller, Category ⟶ partial participation on their side of their relationships (a customer need not have placed and order yet, a seller need not have listed a product yet).

E-R DIAGRAM


Customer
Customer ID
Name
Email
Phone
First Name
Last Name
places
has
order
order date
Contains
Generates
Schedules
Order ID
Order Item
Seller
Category
Payment
Delivery
Address
item NO
Quantity
refers to
Seller ID
Business Name
GST Number
Sells
NAME
Identifies
Category ID
payment ID
Mode
delivers to
Status
Delivery ID
Address ID
full Address
Product
Title
Price
ISA
Product ID
Books
Electronics
fashion
Grocery
Pincode
Street
City
State